

Energy Map – User and initial test plan

Parts of the tool/visualization that should be tested:

1. Ensure that the map and side panels *load* when opened on the touch screen. The *power plants selected* text should read “**34936/34936**”. The year started slider should have the minimum and maximum values “**1896**” and “**2020**” respectively, while the generation slider should have the minimum and maximum values “**0**” and “**82.8k**” respectively. Clicking on the full screen button in the filter panel, should show that most inhabited areas of the world have power plants, to check if all are loaded, filter for “**Zimbabwe**” which has two power plants, one with coal as its primary fuel, and the other with hydro as its primary fuel. Following are the present power plants for each *desired* fuel type (when filtering for said fuel only):
 - a. **Coal** – **2330** power plants
 - b. **Gas** – **3998** power plants
 - c. **Oil** – **2320** power plants
 - d. **Nuclear** – **195** power plants
 - e. **Hydro** – **7156** power plants
 - f. **Wind** – **5344** power plants
 - g. **Solar** – **10665** power plants
 - h. **Other** – **2928** power plants
2. Testing of the different functionalities within the tool, does the toggle feature work in the filter panel? Does the different zoom buttons work?

Some parts of the side panel should also be tested to ensure all functionality works as expected, mainly the navigation buttons at the bottom of the panel should change to the correct page. For the toggleable filters, stored in the drop down menus, the interaction approach should work similarly to the filter panel, i.e., click to select/deselect. If deselection results in all options being deselected, every option will be selected simultaneously. Similarly, if an option is selected while all other options are shown, all options but the selected will be shown. Since the data contains several empty values, such as commissioning year and generation data, the only way to see all of them is to set both slider filters to **[min, max]**. It is expected that power plants with missing data are removed if the dual range sliders are used.

For the pop-up elements on the map, additional clicks on an open element should not open additional elements. Only four pop-ups can be opened at the same time, if additional attempts are made, an alert slides in from the side (might move too quickly, needs to be tested). Can the grey info icon be clicked (hovered if mouse is available) to show what GWh means (sort of). The region information rollup should be openable, which also switches the icon

orientation. Lastly, the bars of the bar chart are clickable as another form to set the fuel filter and if the filter already has been set, there is a small indication (blue border) for the selected fuels.

3. **Readability**, given that the screen size is larger than what the tool was developed on, there is a risk that some text is too small to be readable at a suitable distance, if a problem, it will be fixed. This goes for all text elements present in the tool, but most importantly the instruction text seen in the side panel, both since it guides the user through the tool, and since it already is on the relatively small on the 1920x1080 screen it was developed on. Additionally, it will also be suitable to look over the margins and padding once again, when the text size is evaluated.

How said parts should be tested (user test plan):

Since the user tests are aimed to be conducted on the 54-inch touch displays in Visualisation Center C, it would, according to Carpendale in *Evaluating Information Visualizations*, be suitable to perform the user tests as either *Field Studies* or *Field Experiments* (Carpendale, 2008). Both methodologies aim to be generally unobtrusive compared to *Laboratory Experiments* or *Sample Surveys* for example (McGarth, 1995). However, to ensure that the previously mentioned parts of the tool that should be tested, a *Field Experiment* is more suitable compared to a *Field Study* since it allows for the use of user tasks, which will be described in following paragraphs (Carpendale, 2008). This sacrifice of complete unobtrusiveness aims to reveal possible faults in intended features within the tool.

Tasks:

1. **General use of the tool:** Instruct the test participant to follow the instructions present in the side panel pages. If necessary, tell them that they can click on the icons on the bottom of the page to change the displayed page. If most test participants need this hint, there is a possibility that the visual indicator (highlighted icon for current page) is not enough information to allow for navigation.

Given the instructions present in the pages, most features of the tool will be explored during this task. As mentioned previously, readability will be largely evaluated within the side panel. To ensure that data is collected regarding this, the moderator *should* ask the test participants if they can read and understand the text present for each set of instructions. Additionally, encourage that the test participants describe their actions and thought processes during the test, ensuring to write it down for qualitative data.

2. **Filtering panel functionality:** Instruct the test participant to use the panel in the *bottom-left* side of the tool to filter for all **Coal**, **Gas**, and **Oil** power plants in the world and ask them how many power plants are selected. Correct answer:

8648. Next, ask them to only use the buttons in the panel, along with panning of the map, to zoom into **Belgium** (country does not really matter, but Belgium has many gas power plants). Lastly, ask them to remove all filters and show the entire map, remind them that they can choose any desired method to achieve this. Note down if they used the full screen button and ask *why* they chose the specific method they used.

3. **Data cards:** Instruct the test participants to use the drop-down filters to select all **Hydro** power plants in **Sweden** and task them to find “**Lango** off the coast of Östergötland”. Firstly, this will evaluate if the colour choices are suitable for the current map used since Långö is represented by a rather small green circle. Important that it is logged if the test participant can find the power plant without additional assistance. Whether or not the test participant can find the power plant (if not show where it is), ask them to recite the information they can read about the power plant, which goes as follows:

- a. **Commissioning Year:** 1956
- b. **Primary Fuel:** Hydro
- c. **Capacity:** 1MW
- d. **Generation:** 4.13 GWh

Note down whether the answers are correct or not, next, if they have not done it on their own yet, ask the participant to recite information about power generation in the entirety of Sweden, specifically, **which three primary fuels contribute to most of the generation in Sweden**. Log if the test participants are correct or wrong in their answer. Additionally, keep notes on what the test participants says during this task, especially if it is relevant to the structure of or information in the data cards.

4. **Filtering by drop down menus:** With **Sweden** and **Hydro** still selected in the filter, instruct the test participants to add **Norway** and **Finland**, along with the fuel types **Nuclear** and **Wind**. If they previously have not commented on how satisfactory this process feels, ask them now. If they have any problem with how the drop down menus are structured, it should be noted down. Next, task them to fit all shown power plants into view. Ask which primary fuel is most common among the shown points (correct: **Hydro**) and what region contains the largest concentration of this fuel type (correct: **Norway**). Lastly, ask them which fuel types contributes the most to the actual generation in these areas (correct: **Hydro—Norway, Nuclear—Sweden, Nuclear—Finland**), note down how they found this information.
5. **Filtering by sliders:** Instruct the test participants to reset all filters, i.e., the drop down menu headers show “**All Regions**” and “**All Power Sources**” and finally to reset the map to full screen. The test participants are should now be tasked with finding the oldest power plant in the data (correct: **Granite in the US, 1896**), and how many power plants were commissioned between 1896—1910

(correct: **81**). Instruct them to reset the commissioning year filter and instead find the number of power plants that has an annual generation above 30k GWh (correct: **~26** since the range slider is hard to get exactly on 30k). Additionally, ask them which singular power plant produces the highest amount of electricity in the world (correct: **Kashiwazaki Kariwa in Japan, ~66k**).

6. **Additional comments:** With the functionalities of the tool tested, ask the test participants if they have any additional comments on the tool and note them down if any. Lastly, thank them for their time.
7. **(Possibly, ask the participants to complete a SUS-survey for quantitative data)**